

Zheyi Zhuang

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Education

Carnegie Mellon University, School of Computer Science | Master of Software Engineering December 2026

- Cumulative Grade Point Average: 3.9/4.0
- Relevant Coursework: Quality Assurance, API Design, Software Architectures, Software Management, Devops

University of Washington, Paul G. Allen School of Computer Science | Bachelor of Computer Science December 2022

- Cumulative Grade Point Average: 3.85/4.0; Dean's List: 13 quarters
- Relevant Coursework: Machine Learning, Natural Language Processing, Operating Systems, Algorithms, Computer Graphics

Professional Experience

Microsoft Corporation (Contract) April 2023 - October 2024

Software Engineer I – Team CMD AI Devices

- Built core feature components for Teams Rooms people-identification (C++/C#/Python), supporting deployment across legacy and new conference room hardware.
- Implemented face-enrollment UI flows (TypeScript/Apollo GraphQL/Fluent UI) with privacy and retention checks required for internal approval.
- Automated AI/ML service and E2E pipeline testing, lifting coverage to ~99% and cutting regressions by ~30%.
- Helped optimizing CPU inference paths for 1080p@30fps video, improving latency to ~22ms p95 and ~98% detection accuracy.

BeaconFire Inc., New Jersey (Remote) August 2022 - December 2022

Full-Stack Engineer

- Engineered internal management platform UI (React/Next.js) and onboarding flows so new teams could self-serve setup and configurations, speeding time-to-first-commit from 7 days to 1 day for 50 teams.
- Delivered API layer and persistence (Express + Redis) with auth and audit logging to guard internal workflows, serving 700 internal users at ~99.9% uptime.
- Streamlined CI/CD pipelines (GitHub Actions) to cut release time by ~10 minutes per deploy and stabilize rollback reliability.

Academic Experience

University of Washington, Seattle June 2021 - August 2022

Web Development Teaching Assistant

- Taught 40+ students on advanced web development and algorithms; ran weekly code reviews labs with actionable feedback.
- Started project review checklists and CI gating, increasing assignment pass consistency and code quality.

Projects

[Splendor](#) — Web Board Game

- Developed a real-time Splendor web app in a 4-person team using React/TypeScript, Node/Express, and Socket.IO.
- Designed a domain-driven architecture with command-pattern game logic to enforce rules and keep API contracts clean.
- Created multiplayer features (lobbies, chat, betting, notifications, achievements) backed by SQLite and bcrypt auth.
- Refined reliability with unit, integration, e2e, property-based, and mutation testing (Vitest, Playwright, fast-check, Stryker).

[AlgoExporer](#) — Interactive DSA Learning Platform

- Produced an interactive algorithm visualizer in React/TypeScript, covering 20+ common interview algorithms and data structures with step-by-step animations and multi-language implementation/complexity panels for faster learning.
- Modeled an extensible frame type supporting 10+ data structures to drive playback controls and algorithm-specific UI state.
- Integrated an AI tutor (Gemini + ChatGPT) that injects live visualization context to generate step-specific explanations.

Skills

Languages: TypeScript, JavaScript, Python, C++, C#, Go, Java, SQL, Bash.

Frontend: React, Next.js (App Router), Redux, React Query/SWR, Apollo Client, Tailwind CSS, shadcn/ui, Jest, Playwright.

Backend: GraphQL (Apollo Server), REST, gRPC, WebSocket, Node.js, Express, Java Spring Boot.

Data Storage: PostgreSQL, MongoDB, Redis, Azure Cosmos DB.

Cloud & Infra: Azure (AKS, AAD, Key Vault, App Service, Azure ML, Azure DevOps), AWS (EC2, S3, Lambda), Supabase, Docker.

CI/CD & DevOps: Azure Pipelines, GitHub Actions, Git, Linux/Unix, feature flags.

ML & AI: PyTorch, TensorFlow, HuggingFace, scikit-learn, computer vision, NLP.